Hiring Team
OXMAN
New York, NY

April 29, 2026

Dear Hiring Manager,

I am writing to express my interest in the Machine Learning Research Engineer position for EDEN. Having built systems that model complex spatiotemporal dynamics—most recently using diffusion machine learning to forecast spatial distributions at Homecastr, and engineering high-resolution elevation models from massive-scale LiDAR datasets at Summit Geospatial—I am deeply inspired by OXMAN's commitment to Nature-centric design. As I complete my M.S. in Urban Planning at Columbia GSAPP, my exposure to design thinking has fundamentally shifted how I view computational infrastructure. The opportunity to bridge machine learning with architecture and ecology, synthesizing environmental data to empower ecosystem design, aligns perfectly with my background in spatial modeling and the built environment.

Prior to my current ventures, I spent five years as a Research Engineer and Data Scientist at the Texas Advanced Computing Center (TACC) scaling flood and climate simulations on the world's #1 academic supercomputer, Frontera (TOP500 #5). During my tenure, I led a principal project for the $40M Texas Disaster Information System (TDIS) and partnered with the US Army Corps of Engineers to model compound flood hazards. Modeling complex, non-linear environmental systems at that scale has prepared me for the challenge of integrating generative design with ecosystem simulation models at EDEN.

In reviewing OXMAN's Data-Driven Material Modeling (DDMM) framework and research into direct point-cloud fabrication, I recognize a profound parallel to my work managing high-dimensional spatial data. Whether mapping dynamic environmental point clouds directly to generation instructions—bypassing lossy boundary representations to preserve physical fidelity—or scaling generative models to simulate ecosystem behaviors, I am eager to apply my computational expertise to your integrated design workflows. I would welcome a conversation to discuss how my background in spatial ML can contribute to EDEN's vision of cohabitation with Nature.

Sincerely,

Daniel Hardesty Lewis

Attached: curriculum vitæ

Daniel Hardesty Lewis
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Daniel Hardesty Lewis

Machine Learning Research Engineer – geospatial ML, ecosystem design, distributed infrastructure

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

Core Expertise

8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs

10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT

LLM apps embeddings; RAG; LangChain; LangGraph; DSPy; FAISS

Data stack Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase

Infrastructure Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability

Additional CUDA; C++; TypeScript; React

Experience

Professional

Nov. 2024 – Present **Co-founder**, *Homecastr*, New York, NY

Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.

- Architected a nationwide neighborhood-level diffusion forecasting model, **matching Zillow's 7.3% 0-year error benchmark at a 1-year forecast horizon** and maintaining a **35.6% median error over 4 years**.
- Engineered architecture incorporating **spatial inducing tokens**, **per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
- Developed predictive validation optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
- Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z and MetaProp**.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow for BOM parsing and tariff simulation using Airflow, Elasticsearch embeddings, and FastAPI on Supabase and Redis.
- Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
- Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing $1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

Research

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
- Math benchmarking and finetuning of Deepseek's GRPO algorithm

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX
Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

M&A predictor Built M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr.

Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

Languages

Spanish Native Proficiency

French Professional Working Proficiency

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